Worksheet 6-3 A Definite Integral Notation #1—ANSWERS
AP Calculus

1.) $\int_0^1 \frac{e^x}{1+x}\,dx$

2.) $\int_2^7 (5x^3-4x)\,dx$

3.) $\int_0^{\pi} \cos x\,dx$

4.) $\int_{-3}^{2} \sqrt[3]{x}\,dx$

5.) $\int_6^{15} \left(\frac{1}{x^2}\right) dx$

6.) $\int_0^{\frac{\pi}{4}} \tan x\,dx$

7.) $\int_{-1}^{7} 4x\,dx$

8.) $\lim_{n\to\infty} \sum_{k=1}^{n} \left[-1+\frac{4k}{n}\right]^2 \cdot \left(\frac{4}{n}\right)$

9.) $\lim_{n\to\infty} \sum_{k=1}^{n} \left[\frac{1}{2+\frac{3k}{n}}\right] \cdot \frac{3}{n}$

10.) $\lim_{n\to\infty} \sum_{k=1}^{n} \ln\left(3+\frac{k}{n}\right) \cdot \frac{1}{n}$

11.) $\lim_{n\to\infty} \sum_{k=1}^{n} \sqrt{4+\left(1+\frac{2k}{n}\right)^2} \cdot \frac{2}{n}$

12.) $\lim_{n\to\infty} \sum_{k=1}^{n} \left[\left(2+\frac{3k}{n}\right)^2 + \frac{1}{2+\frac{3k}{n}}\right] \cdot \frac{3}{n}$